

Qatar-9b

R-2493

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250117 · created 2026-08-24T01:31:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460693.012 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.205 +/- 0.06
Transit depth (Rp/Rs)^2	4.22 +/- 2.45 [%]
Orbital Inclination (inc)	89.92 +/- 1.67
Airmass coefficient 1 (a1)	0.1359 +/- 0.0085
Airmass coefficient 2 (a2)	1.75 +/- 0.64
Scatter in the residuals of the lightcurve fit is	6.37 %
Best Comparison Star	#1 - [340, 246]
Optimal Aperture	1.88
Optimal Annulus	10.74
Transit Duration (day)	0.029 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.205 ± 0.06 vs 0.1489 (deviation 0.7σ)
Duration fit vs published	0.029 d vs 0.0779 d (deviation 3.35σ)
Mid-time O-C	21.3 min
Depth SNR	2.6
Residual scatter	6.37 %

Light curve

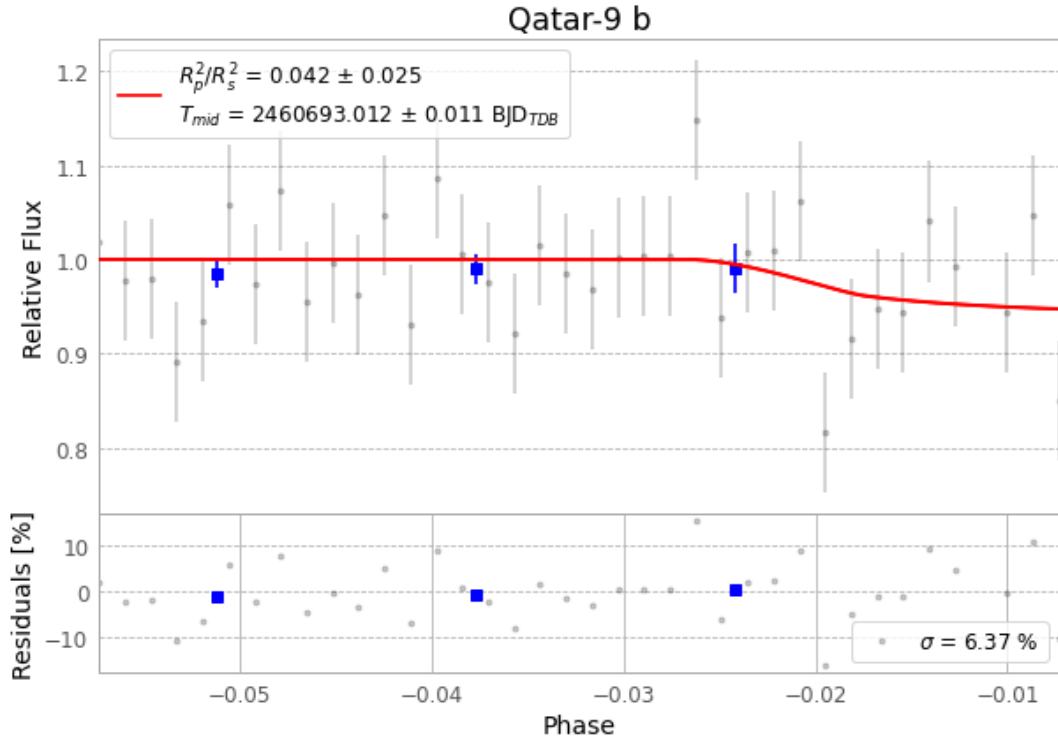


Figure 1 — FinalLightCurve_Qatar-9 b_2025-01-17.png (SHA-256 d43a33fbf6d95a04...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 277], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df29b944283deb42...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

38 FITS frames (25 MB), Qatar-9250117100315.FITS → Qatar-9250117115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250117.log (5875733bbb2c...), FinalLightCurve_Qatar-9 b_2025-01-17.png (d43a33fbf6d9...), FinalLightCurve_Qatar-9 b_2025-01-17.csv (730632c11ee3...)

Compute resources

Wall time 97.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 01:31Z.